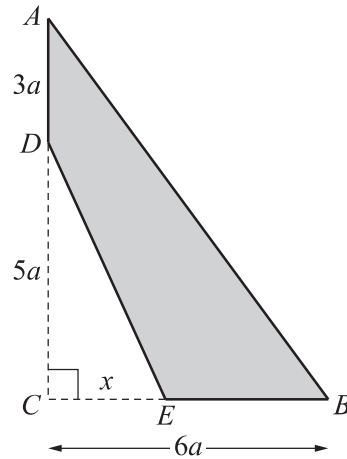




A uniform lamina is in the form of a triangle ABC , with $AC = 8a$, $BC = 6a$ and angle $ACB = 90^\circ$. The point D on AC is such that $AD = 3a$. The point E on CB is such that $CE = x$ (see diagram). The triangle CDE is removed from the lamina.

- (a) Find, in terms of a and x , the distance of the centre of mass of the remaining object $ADEB$ from AC . [4]

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice. There are no margins, text, or other markings on the page.

The object $ADEB$ is on the point of toppling about the point E when the object is in the vertical plane with its edge EB on a smooth horizontal surface.

- (b) Find x in terms of a . [3]

This image shows a full page of a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.